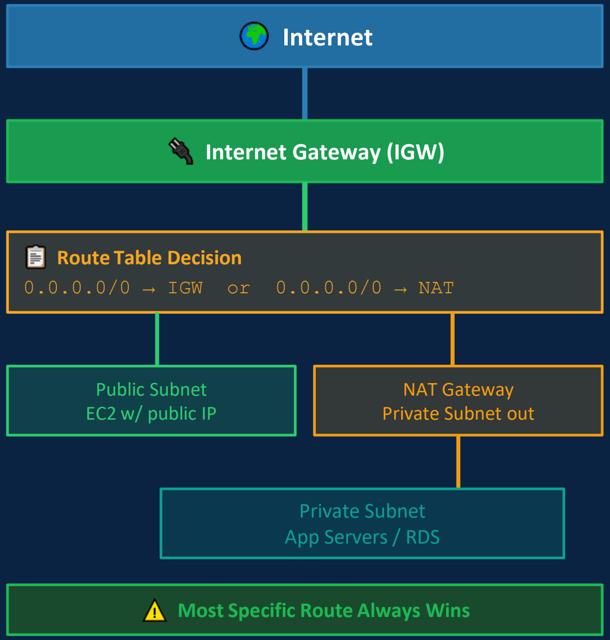
WEEK 3 · DAY 3

Route Tables,

Internet Gateways & NAT

BaseStack Academy · AWS Cloud Accelerator

Networking & VPC



Cohort 1 | Beginner-Friendly | SAA-C03 Aligned

basestack.academy

Today's Agenda

1. **Overview — Traffic Control Is the Heart of VPC Security**
   FOUNDATION
2. **Key Concepts — Route Tables, Entries, Gateways**
   CORE KNOWLEDGE
3. **Route Table Deep Dive — Local, Priority Rules & Multiple Tables**
   DEEPER LEARNING
4. **How It Works — Creating & Associating Route Tables**
   MECHANICS
5. **Technical Deep Dive — Traffic Flow Tracing & NAT Cost Model**
   ADVANCED DETAIL
6. **Architecture & Visual Model — Full Routing Pipeline**
   VISUAL
7. **Real-World Nigerian Scenario**
   CONTEXT
8. **Common Mistakes & Exam Traps**
   EXAM PREP

📖 Overview — Traffic Control Is the Heart of VPC Security

On Day 1 we built the VPC. On Day 2 we designed the subnets. Today we answer the question those two days left open: how does network traffic actually know where to go?

The answer is route tables. Every packet that enters, leaves, or travels within your VPC is evaluated against a route table. The route table matches the destination IP address of the packet against its list of routes and decides: send this packet here. Route tables are the traffic control system of your entire cloud network — the rules that determine which subnets have internet access, which resources can reach each other, and which traffic paths are completely blocked.

Without understanding route tables, you cannot reason about why traffic flows or why it does not. You cannot debug connectivity problems. You cannot design secure network segmentation. Every VPC troubleshooting conversation eventually comes down to: what does the route table say?

WHY NOW?

Route table design directly determines your compliance posture. A route table with 0.0.0.0/0 → IGW on a database subnet means that database has an internet path — a PCI-DSS and CBN violation. A route table with only the local VPC route means the database is provably isolated. The route table IS the network security evidence.

SAA-C03 Domain 1 and Domain 2 test route table design extensively — which targets to use for which scenarios, how priority resolution works, and what happens when a subnet has no explicit route table association.

In interviews: 'walk me through how traffic flows from a user to your RDS database' is a route table question in disguise.

💡 **Beginner Note**

Think of a route table as the GPS navigation system for your VPC. Every packet asks 'where should I go?' and the route table answers. The route table does not carry the traffic — it just tells traffic which road to take.

🔑 Key Concepts — Route Tables, Gateways & Entries

🎓 *Every concept here has a direct impact on traffic flow. Know the mechanism, not just the name.*

Route Table

A set of rules (routes) that determine where network traffic from a subnet is directed. Each route has two parts: a Destination (a CIDR block like 10.0.0.0/16 or 0.0.0.0/0) and a Target (where to send matching traffic — e.g., igw-xxx, nat-xxx, or 'local'). A subnet can be associated with exactly ONE route table at a time.

Internet Gateway (IGW)

A horizontally-scaled, redundant, AWS-managed gateway attached to your VPC that enables bidirectional communication between the VPC and the public internet. A subnet becomes 'public' when its route table has 0.0.0.0/0 → igw-xxx. The IGW also performs Network Address Translation (NAT) — converting EC2 private IPs to public IPs for outbound traffic.

Route Entry Anatomy

Every route has: Destination CIDR — the IP range this rule applies to. Target — where matching traffic is sent. When a packet arrives, AWS checks ALL routes and selects the MOST SPECIFIC match (longest prefix). 10.0.1.0/24 beats 10.0.0.0/16 beats 0.0.0.0/0 for a packet going to 10.0.1.5.

NAT Gateway

A managed service that sits in a PUBLIC subnet and allows resources in PRIVATE subnets to initiate OUTBOUND connections to the internet while blocking all UNSOLICITED INBOUND traffic. Requires an Elastic IP. Costs: $0.045/hour + $0.045/GB processed. For HA, deploy one per AZ — cross-AZ NAT traffic incurs extra data transfer charges.

Local Route

The automatic, undeletable route that every route table contains: Destination 10.0.0.0/16 (your VPC CIDR) → Target: local. This route means all traffic destined for any address within the VPC stays inside the VPC and never leaves. You cannot delete or modify this route.

Egress-Only Internet Gateway

Like a NAT Gateway but for IPv6 traffic only. Since IPv6 addresses are globally unique (no NAT), a regular IGW would allow both inbound and outbound IPv6 traffic. An Egress-Only IGW allows outbound IPv6 from private resources while blocking unsolicited inbound IPv6. Stateful — automatically allows return traffic.

🔑 Route Table Deep Dive — Priority, Local Route & Multiple Tables

Three things about route tables that are almost never properly explained — and are always exam-tested.

⚡ Rule 1: The Most Specific Route Always Wins

When a packet's destination matches multiple routes, AWS uses the LONGEST PREFIX MATCH — the most specific (narrowest) route wins.

10.0.1.0/24	local	← WINS for 10.0.1.5 (most specific /24)
10.0.0.0/16	local	← loses (less specific /16)
0.0.0.0/0	igw-xxx	← loses (least specific — catch-all)

🔒 Rule 2: The Local Route Is Sacred — Never Removable

Every route table has exactly one local route automatically added at VPC creation. You CANNOT delete it, modify it, or override it. It always routes VPC-internal traffic locally.

10.0.0.0/16	local	ALWAYS present. Cannot be removed.

This means: a packet from 10.0.1.5 to 10.0.21.8 (your RDS) ALWAYS stays inside the VPC — the local route intercepts it before 0.0.0.0/0 could send it to the internet.

📋 Rule 3: One VPC Can Have Multiple Route Tables

The Main Route Table is created automatically with your VPC. All subnets that are NOT explicitly associated with a custom route table use the Main Route Table by default.

main-rt (Main)	public-rt	private-rt	isolated-rt
Unassociated subnets	prod-public-az1a, az1b	prod-app-az1a, az1b	prod-data-az1a, az1b
`10.0.0.0/16 → local`	`10.0.0.0/16 → local` `0.0.0.0/0 → igw-xxx`	`10.0.0.0/16 → local` `0.0.0.0/0 → nat-xxx`	`10.0.0.0/16 → local ONLY`

⚠️ Rule 4: Subnet Without Association → Uses Main Route Table

This is a dangerous default. If you create a subnet and forget to associate it with your custom route table, it uses the Main Route Table instead.

Best practice: Keep the Main Route Table as LOCAL ONLY. Never add an IGW or NAT route to the Main Route Table. Explicitly associate every subnet with a named custom route table. This way, a forgotten association is safe — it just gets the local-only table — rather than accidentally getting internet access.

🔴 Anti-pattern: Adding 0.0.0.0/0 → IGW to the Main Route Table. Every new subnet you create would automatically get internet access — including your database subnets — until you remember to associate them with the correct table.

How It Works — Creating & Associating Route Tables

1 Create Custom Route Tables (One Per Traffic Pattern)

Navigate to VPC → Route Tables → Create Route Table. Create three tables: public-rt, private-rt, and isolated-rt. Name them descriptively — the name appears in subnet association views and saves confusion at 2am during an incident. At creation, each table has only one route: the automatic local route for your VPC CIDR.

Best practice: Immediately tag each route table with the Environment, Tier, and Owner. This is your future operational metadata.

2 Add Routes to Each Table

For public-rt: click Routes tab → Edit routes → Add route. Destination: 0.0.0.0/0. Target: select Internet Gateway → select your IGW. Save. This is the route that makes subnets associated with this table 'public.'

For private-rt: Add route. Destination: 0.0.0.0/0. Target: NAT Gateway → select your NAT GW. Save. Private subnets using this table get outbound-only internet via NAT.

For isolated-rt: Add NO additional routes. The local VPC route is sufficient. No internet path in either direction.

3 Associate Subnets to Route Tables

Go to each route table → Subnet Associations tab → Edit subnet associations. Select your public subnets for public-rt. Select your private-app subnets for private-rt. Select your private-data subnets for isolated-rt.

VERIFY: Go to each subnet → Route Table tab. Confirm the association. Confirm the routes shown are exactly what you expect. A public subnet should show 0.0.0.0/0 → igw-xxx. A data subnet should show ONLY the local VPC route. If anything else appears, fix it before deploying resources.

Technical Deep Dive — Traffic Flow Tracing & NAT Cost Model

End-to-End Traffic Flow Trace

1 User → ALB (Inbound Public)

User IP	→ ALB public IP	Internet → IGW → Public Subnet
ALB SG	443 allowed	SG permits HTTPS from 0.0.0.0/0

2 ALB → EC2 App (Internal)

ALB	→ EC2 10.0.11.x	Internal VPC traffic via local route
Route	10.0.0.0/16 → local	No internet involved — VPC-internal

3 EC2 App → RDS (Internal)

EC2	→ RDS 10.0.21.x	Both in same VPC — local route
Route	10.0.0.0/16 → local	Isolated-rt: NO other routes

4 EC2 App → Internet (Outbound via NAT)

EC2	→ api.paystack.co	Private subnet outbound request
Route	0.0.0.0/0 → nat-xxx	Private-rt sends to NAT Gateway

NAT Gateway — Cost Model & HA Design

💰 NAT Gateway Pricing (eu-west-1)

Per hour (whether used or not)	$0.045/hr ≈ $32.85/month
Per GB of data processed	$0.045/GB
Cross-AZ data transfer	Extra $0.01/GB
1 NAT GW × 1 month + 100GB/day	≈ $166/month

🌐 NAT Gateway HA — One Per AZ

NAT Gateways are AZ-scoped — they live in ONE AZ. If that AZ fails, the NAT GW fails too.

For HA: Deploy one NAT GW per AZ. Each AZ's private subnets route to THEIR OWN NAT GW.

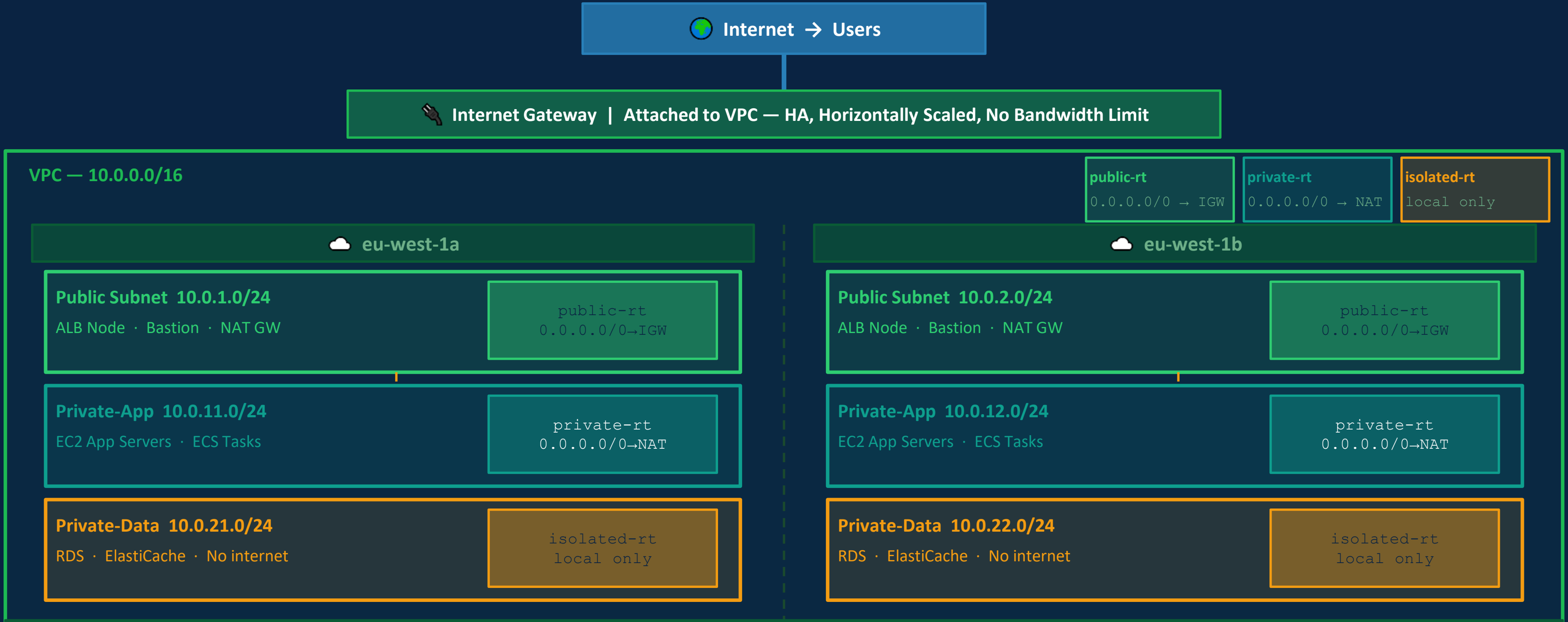
⚖️ NAT Gateway vs NAT Instance

NAT Gateway: Managed, HA, scalable, no patching, higher cost

NAT Instance: EC2-based, manual HA, you manage patches, lower cost at very low traffic

🏗️ Architecture — Complete Routing Pipeline Across 2 AZs

Full end-to-end routing: three route tables, two Internet Gateway routes, two NAT Gateways (one per AZ for HA), and an isolated data tier with zero internet routing.



Each AZ has its own NAT GW (HA) · IGW is shared, VPC-level · isolated-rt has NO 0.0.0.0/0 entry · 'Most specific route wins' resolves all conflicts

🌍 Real-World Nigerian Scenario — OPay-Style Digital Payment Platform

SCENARIO: A Nigerian digital payment platform — modelled on OPay's architecture — serves 35 million registered users with services including money transfers, bill payments, and merchant QR payments. Their DevOps team is redesigning the routing architecture after a security audit flagged incorrect route table configurations that accidentally exposed internal services. The platform processes CBN-regulated financial transactions and must demonstrate network isolation to pass their upcoming IT governance audit.

🔍 Route Table Audit Findings & Remediation

🔴 FINDING 1: Main Route Table Had 0.0.0.0/0 → IGW

The team discovered that 4 subnets — including two hosting internal microservices — were not explicitly associated with any route table and defaulted to the Main Route Table. Because someone had added an IGW route to the Main Route Table 'temporarily' during debugging 6 months ago, those 4 subnets were publicly routable. The internal payment processing API was reachable from the internet. The security audit flagged this as a critical finding.

✅ **FIX:**

REMEDIATION: Removed the IGW route from the Main Route Table. The Main Route Table now contains ONLY the local VPC route. Created explicit route tables for all subnets. Associated every subnet explicitly — no subnet is allowed to fall back to Main.

🟡 FINDING 2: Single NAT Gateway Creating an AZ Single Point of Failure

The platform had one NAT Gateway in eu-west-1a serving private subnets in BOTH eu-west-1a and eu-west-1b. The private-rt for eu-west-1b pointed 0.0.0.0/0 to a NAT GW in a different AZ. During a brief AZ-1a connectivity issue, ALL outbound internet traffic from eu-west-1b private subnets failed — including the payment processing services that call external card networks.

✅ **FIX:**

REMEDIATION: Deployed a second NAT Gateway in eu-west-1b. Created separate private route tables per AZ: private-rt-az1a routes to nat-az1a. private-rt-az1b routes to nat-az1b. Each AZ is now independently resilient. Additional cost: ~$32/month for the second NAT GW — acceptable for a platform processing millions of transactions.

🟢 FINDING 3: No Route Table Documentation — Audit Evidence Gap

The CBN IT governance audit requires documented evidence of network segmentation. The team could not produce clean documentation showing which subnets were isolated from the internet, because route tables had no consistent naming or tagging, and subnet associations were difficult to trace from the console alone.

✅ **FIX:**

REMEDIATION: Implemented IaC (Terraform) for all route tables with mandatory tags: Name, Environment, Tier, Owner, LastReviewed. Automated compliance report using AWS Config Rule that checks route tables for unexpected 0.0.0.0/0 routes on data tier subnets. Alert fires within minutes if someone adds an internet route to an isolated subnet.

💬 Discussion: Why is an AWS Config Rule a stronger compliance control than a NACL for detecting unexpected internet routes on isolated subnets?

⚠ Common Mistakes & Exam Traps

These are the most common SAA-C03 exam errors on Route Tables & Gateways — every one costs marks.

✗ **TRAP 1: NAT Gateway is NOT free — it charges per hour AND per GB. For HA, you need one per AZ.**

A common exam distractor states that NAT Gateway is free within the same Region. It is NOT free. You pay $0.045 per hour per NAT Gateway (approximately $32.85/month even if it processes zero traffic) plus $0.045 per GB of data processed. For High Availability across two AZs, you need two NAT Gateways — double the hourly cost. Exam questions about cost-optimisation for low-traffic private subnet workloads often point to NAT Instance as the cheaper alternative.

✗ **TRAP 2: The Internet Gateway must be ATTACHED to the VPC AND the subnet's route table must have a route pointing to it.**

Creating an IGW does nothing until you attach it to the VPC. Attaching it to the VPC does nothing until the subnet's route table has 0.0.0.0/0 → igw-xxx. AND the EC2 instance needs a public or Elastic IP. ALL THREE must be true simultaneously for internet connectivity. Exam scenarios where traffic isn't flowing often have exactly one of these three missing — identify which one.

✗ **TRAP 3: NAT Gateway only supports TCP, UDP, and ICMP — not all protocols.**

NAT Gateway does NOT support GRE (used for VPN tunnels like PPTP), IPsec ESP/AH packets in transport mode, or other non-TCP/UDP/ICMP protocols. Exam questions sometimes describe a scenario where a specific protocol is failing through a NAT Gateway — if the protocol is GRE or IPsec, the NAT Gateway is the reason, and the solution requires a VPN Gateway or a NAT Instance with IP forwarding enabled, not another NAT Gateway.

✗ **TRAP 4: A subnet with NO explicit route table association uses the Main Route Table — not 'no routing'.**

This trips up many candidates. If you create a subnet and don't associate it with any route table, it does NOT have no routing — it automatically inherits the Main Route Table for the VPC. If someone has added internet routes to the Main Route Table, that unassociated subnet gets internet access. Best practice: keep the Main Route Table as local-only and explicitly associate every subnet with a named custom table.

✗ **TRAP 5: The local VPC route (10.0.0.0/16 → local) is permanent — it always takes precedence over other routes for VPC-internal traffic.**

You cannot delete or modify the local route. You cannot add a route that overrides it for VPC-internal traffic — any destination that falls within the VPC CIDR will ALWAYS be routed locally, even if you add a 0.0.0.0/0 route. This means traffic between a public subnet and a private subnet NEVER goes through the IGW — it always takes the local path. Many candidates mistakenly think the IGW handles internal VPC routing.

Question

A solutions architect is reviewing the VPC configuration for a Lagos-based fintech. The VPC has three subnets: a public subnet hosting an Application Load Balancer, a private subnet hosting EC2 application servers, and a private subnet hosting an RDS database. The EC2 application servers must be able to download security patches from the internet. The RDS database must have NO internet connectivity in any direction. Currently, all three subnets are associated with the Main Route Table which has the entry: 0.0.0.0/0 → igw-xxxxxxxx.

Which set of actions will correctly implement the required routing? (Select THREE)

A — **Create a NAT Gateway in the public subnet. Create a custom route table with routes: 10.0.0.0/16 → local and 0.0.0.0/0 → NAT Gateway. Associate this route table with the private EC2 subnet.** ✓ CORRECT

Correct. A NAT Gateway in the public subnet enables outbound-only internet for the EC2 app servers. The private route table with the NAT GW route allows patch downloads without exposing instances to inbound connections.

B — **Create a custom route table with only the local VPC route (10.0.0.0/16 → local). Associate this route table with the RDS subnet.** ✓ CORRECT

Correct. An isolated route table with only the local route ensures the RDS subnet has NO path to the internet — not even outbound through a NAT Gateway. This satisfies the zero-internet-connectivity requirement for the database.

C — **Remove the 0.0.0.0/0 → igw-xxxxxxxx route from the Main Route Table. Create a custom public route table with 10.0.0.0/16 → local and 0.0.0.0/0 → igw-xxxxxxxx. Associate it with the ALB subnet.** ✓ CORRECT

Correct. The Main Route Table should not have the IGW route — it affects all unassociated subnets. Removing it and creating a dedicated public-rt associated only with the ALB subnet implements proper routing separation.

D — Add a Security Group rule to the RDS instance denying all inbound traffic from 0.0.0.0/0. Keep the existing Main Route Table association on the RDS subnet.

Wrong. A Security Group is an operational control, not a network routing control. The RDS subnet would still have a route to the internet via the IGW. The requirement is NO internet connectivity — enforced at the route table (network) layer, not the Security Group (application) layer.

E — Create an Egress-Only Internet Gateway and associate it with the RDS subnet to allow outbound-only access for database updates.

Wrong. An Egress-Only IGW is for IPv6 outbound-only traffic. It is not appropriate here. More fundamentally, the requirement states NO internet connectivity in any direction — including outbound. The RDS subnet should have no internet route of any type.

Key Takeaways — What You Learned Today

Route Tables, Internet Gateways & NAT

Concepts Covered

- ✓ Route table = destination CIDR + target. Most specific route wins.
- ✓ Local route (VPC CIDR → local) is permanent and undeletable.
- ✓ IGW enables bidirectional internet. Subnet = public when RT has 0.0.0.0/0 → IGW.
- ✓ NAT GW: outbound-only for private subnets. NOT free ($0.045/hr + $0.045/GB).
- ✓ For HA: one NAT Gateway per AZ with separate per-AZ private route tables.
- ✓ Main Route Table is the default for unassociated subnets — keep it local-only.
- ✓ Subnets with no explicit RT association inherit the Main Route Table.
- ✓ Egress-Only IGW: IPv6 outbound-only. Not for IPv4. Not for isolated data tiers.
- ✓ NAT GW supports TCP/UDP/ICMP only — not GRE or raw IPsec protocols.
- ✓ Route table is compliance evidence — not Security Group, not NACL.

Before Next Class

1. Recall all 10 concepts above without notes — active retrieval first
2. Console lab: create public-rt, private-rt, isolated-rt. Add appropriate routes to each.
3. Associate your subnets from Day 2's lab to their correct route tables
4. Verify: Public subnet shows IGW route. Private shows NAT route. Data shows local only.
5. Trace a full request from browser → ALB → EC2 → RDS and back. Draw each hop.
6. Calculate: what would 2 NAT Gateways + 500GB/month data cost? (Answer on WA group)
7. Check your Main Route Table — confirm it has ONLY the local VPC route

Further Resources & Reading

Go deeper with these official AWS and industry resources:

VPC Route Tables — Official Documentation

OFFICIAL DOCS

https://docs.aws.amazon.com/vpc/latest/userguide/VPC_Route_Tables.html
Complete reference covering route entry configuration, Main Route Table behaviour, subnet associations, route priority resolution, and route propagation for VPN/Direct Connect. The 'Route priority' section is essential exam reading.

Internet Gateways — AWS Documentation

OFFICIAL DOCS

https://docs.aws.amazon.com/vpc/latest/userguide/VPC_Internet_Gateway.html
How IGWs work, attachment requirements, NAT behaviour for public IPs, the three conditions for internet connectivity (IGW attached + route + public IP), and limitations. Covers the difference between an auto-assigned public IP and an Elastic IP.

NAT Gateways — Configuration & Pricing

PRICING + OPS

https://docs.aws.amazon.com/vpc/latest/userguide/vpc-nat-gateway.html
NAT Gateway setup, Elastic IP requirements, multi-AZ deployment pattern, bandwidth limits (up to 100 Gbps), protocol support (TCP/UDP/ICMP only), and pricing model. Read the 'NAT gateway use cases' section for exam scenarios on when to use NAT Gateway vs NAT Instance.

Egress-Only Internet Gateway — IPv6 Outbound

ADVANCED

https://docs.aws.amazon.com/vpc/latest/userguide/egress-only-internet-gateway.html
Why IPv6 needs a separate gateway mechanism (no NAT in IPv6 — all addresses are globally unique), how Egress-Only IGW provides outbound-only IPv6 for private resources, and the SAA-C03 exam context where this appears (IPv6-enabled VPCs).

SAA-C03 Domain 1 — Resilient Network Architecture

EXAM PREP

https://d1.awsstatic.com/training-and-certification/docs-sa-assoc/AWS-Certified-Solutions-Architect-Associate_Exam-Guide.pdf
Domain 1 covers multi-AZ routing, NAT Gateway HA patterns, and route table design for resilience. Domain 2 covers routing as a security control — specifically isolated route tables for compliance. Both domains test route table configuration extensively.

Homework: Read at least ONE resource above before next class. Post what you learned on the WhatsApp group.

BASE STACK

ACADEMY

Week 3 · Day 3 Complete

Route Tables, Internet Gateways & NAT

Networking & VPC

Next: Hands-on Console Lab →